40 From excavation data to public narrative: Artificial Intelligence as a bridge in the communication of archaeological stratigraphy

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Within the framework of the digital transition and the enhancement of archaeological heritage, communicating archaeological stratigraphy presents a complex methodological challenge: translating the excavation sequence into an accessible narrative for the general public without compromising scientific rigor. This paper explores the potential of Artificial Intelligence (AI) as a mediator of complex archaeological content for broader audiences. Often perceived as a jargon accessible only to specialists, archaeological stratigraphy embodies a dynamic narrative of time and human transformations that deserves to be conveyed to the wider community with scientific accuracy.

This contribution presents the development and architecture of an interactive web-app prototype that leverages Artificial Intelligence to make stratigraphy understandable and engaging for a nonspecialist public. To critically mitigate the risk of algorithmic hallucinations, the prototype is grounded in an open data infrastructure rigorously validated by archaeologists. Through an intuitive, multi-scale interface, the application processes relational and chronological excavation data, serving as a tangible bridge between specialized vocabulary and public storytelling. Users are guided through a dynamic exploration of historical phases and stratigraphic sequence via chronological reconstructions, evolving from passive consumers into active participants in a shared cultural experience.

The paper discusses the methodological choices, open-source design constraints, and potential applications of the prototype in museums, cultural tourism, and advanced education. In particular, it highlights how the responsible use of AI supports researchers in redefining their role: not only guardians of scientific data, but facilitators of innovative pathways capable of enhancing heritage accessibility and lowering cognitive barriers between scientific research and society.